

Linear systems and elimination

Exponential smoothed tail bounds for partial pivoting

Difficulty: extreme **Importance:** broadly interesting**Status:** [Lean verified](#)

Rating rationale: Extreme reflects a uniform tail bound over adversarial matrix centers, beyond current average-case analysis; broad impact is justified by explaining the stability of a standard dense solver.

Resolution — 11 September 2026

Solved negatively. George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA, gives a counterexample to the displayed universal exponential tail. The [complete proof's theorem](#), [equation \(2\)](#), and [robustness lemma](#) show that, for every $n \geq 2$,

$$\Pr\{\rho_{\text{PP}}(I_n + G) > \frac{1}{2}(3/2)^{n-1}\} \geq 2^{-n^2(n^2+n+5)}.$$

For any proposed $c_1, c_2 > 0$, take $\bar{A} = I_n$, $\sigma = 1$ and $x = (3/2)^{n-1}/(2n^{c_1})$. Sufficiently large n give an admissible $x \geq 1$ and violate the proposed upper bound. This covers the exact real Gaussian model and every proposed universal pair; the event has strict pivot choices and nonsingular matrices. The conclusion uses the unrestricted range of x and does not address a different tail restricted to smaller x .

[Proof PDF](#) · [Standalone source](#) · [Independent complete-proof review](#) · [Submission and verification record](#). AI assistance and automated-review limits are disclosed. The permanent ID, original statement, and historical ratings below are retained.

Lean verification - 15 September 2026

Formalization: Sidney Holden, with OpenAI Codex assistance. George Stepaniants retains authorship of the mathematical argument; Spielman and Teng retain attribution for the original conjecture.

The complete original universally quantified exponential-tail conjecture is formally refuted. The proof uses actual GEPP states and every legal largest-magnitude pivot schedule, their finite growth maxima, the Euclidean operator norm, and the full independent real Gaussian matrix law. The identity-centered counterexample violates every proposed positive pair of constants for sufficiently large admissible dimension and threshold. The full entrywise robustness box and its Gaussian probability lower bound are proved; no sampled-grid substitute is used. The manuscript's additional zero-centered result is not claimed here.

All twelve reviewed exports passed LeanCert kernel trust audits, standard-axiom checks, and [fresh isolated Linux Comparator/default-kernel verification](#) at the [immutable proof revision](#). Two independent statement reviews preceded proofs, and two fresh independent nonauthor final source reviews passed. Their evidence-packaging follow-up is resolved in the retained final code gate.

[Proof, reviews and metadata](#) · [Retained Linux evidence](#). Actual rejection and sandbox controls passed. These are AI-agent reviews, not human peer review, official Tau Ceti endorsement or source-author endorsement. The permanent ID, original target and historical ratings remain unchanged.

Context and notation

All elimination in this problem is in exact arithmetic. Write $\|A\|_{\max} = \max_{ij} |a_{ij}|$. A pivoting path creates successive active Schur complements $S_1 = A, S_2, \dots, S_n$, with row/column permutations as appropriate. Its element-growth factor is

$$\rho(A) = \frac{\max_{1 \leq j \leq n} \|S_j\|_{\max}}{\|A\|_{\max}}.$$

Partial pivoting chooses a largest-magnitude entry in the active first column; complete pivoting chooses one anywhere in the active matrix. If ties occur, a universal statement includes every admissible tie choice; a supremum includes all admissible paths. These conventions remove implementation-dependent ambiguity. Matrices are nonsingular unless otherwise stated.

Problem statement

Do universal constants $c_1, c_2 > 0$ exist such that, for every $n \geq 1$, deterministic $\bar{A} \in \mathbb{R}^{n \times n}$ with $\|\bar{A}\|_2 \leq 1$, $0 < \sigma \leq 1$, and $x \geq 1$, the matrix $A = \bar{A} + \sigma G$, with independent standard normal entries in G , satisfies

$$\Pr\{\rho_{\text{PP}}(A) > x(n/\sigma)^{c_1}\} \leq 2^{-c_2 x}?$$

This is a uniform smoothed-analysis question: the deterministic center may itself be a worst-case input. Almost surely the perturbed matrix is nonsingular and the pivot choices have no ties. Numerical evidence or estimates only at $\bar{A} = 0$ do not settle the quantifier over deterministic centers. The tail estimate would quantify the rarity of substantial growth after small Gaussian input perturbations.

References

Spielman and Teng, *Smoothed Analysis of Algorithms and Heuristics: Progress and Open Questions*, §P6, Conjecture 16, p. 52 of the author PDF; in *Foundations of Computational Mathematics, Santander 2005* (2006), 274–342, [chapter DOI](#). Huang and Tikhomirov, *Average-case analysis of the Gaussian elimination with partial pivoting*, PTRF 189 (2024), introduction.

Earlier status check — 2026-09-08

Searches for *Exponential Stability of GEPP solved* and *Gaussian elimination smoothed analysis* 2025 2026 found the average-case theorem, not this exponential tail bound. The 2026 Peca-Medlin butterfly paper also distinguishes average-case results from the still-unavailable full smoothed analysis. Randomizing the pivot rule is a different model.

Audit update — 2026-09-10

The linked Spielman–Teng chapter was checked against the [author bibliography](#): its title has been corrected in the reference, with Conjecture 16 unchanged. [Huang–Tikhomirov](#) explicitly discusses the obstruction to extending its Gaussian result to arbitrary centers. Searches for exponential GEPP tails and subsequent smoothed-analysis results found no theorem with the displayed uniform quantifiers.